

APPENDIX D

BUILDINGS EXEMPTED FROM TORSIONAL WIND LOAD CASES

D.1 SCOPE

The torsional load cases in Fig. 27.3-8 (Case 2 and Case 4) need not be considered for a building meeting the conditions of Sections D.2, D.3, D.4, D.5, or D.6 or, if it can be shown by other means that the torsional load cases of Fig. 27.3-8 do not control the design.

D.2 ONE- AND TWO-STORY BUILDINGS MEETING THE FOLLOWING REQUIREMENTS

One-story buildings with h less than or equal to 30 ft (9.2 m), buildings two stories or fewer framed with light-frame construction, and buildings two stories or fewer designed with flexible diaphragms are exempted.

D.3 BUILDINGS CONTROLLED BY SEISMIC LOADING

D.3.1 Buildings with Diaphragms at Each Level That Are Not Flexible. Building structures are exempted that are regular under seismic load (as defined in Section 12.3.2) and conform to the following:

1. The eccentricity between the center of mass and the geometric centroid of the building at that level shall not exceed 15% of the overall building width along each principal axis considered at each level, and
2. The design story shear determined for earthquake load as specified in Chapter 12 at each floor level shall be at least 1.5 times the design story shear determined for wind loads as specified herein.

The design earthquake and wind load cases considered when evaluating this exception shall be the load cases without torsion.

D.3.2 Buildings with Diaphragms at Each Level That Are Flexible. Building structures are exempted that are regular under seismic load (as defined in Section 12.3.2) and conform to the following:

1. The design earthquake shear forces resolved to the vertical elements of the lateral load-resisting system shall be at least 1.5 times the corresponding design wind shear forces resisted by those elements.

The design earthquake and wind load cases considered when evaluating this exception shall be the load cases without torsion.

D.4 BUILDINGS CLASSIFIED AS TORSIONALLY REGULAR UNDER WIND LOAD

Buildings meeting the definition of torsionally regular buildings under wind load contained in Section 26.2 are exempted.

EXCEPTION: If a building does not qualify as being torsionally regular under wind load, it is permissible to base the design on the basic wind load Case 1 that is proportionally increased so that the maximum displacement at each level is not less than the maximum displacement for the torsional load Case 2.

D.5 BUILDINGS WITH DIAPHRAGMS THAT ARE FLEXIBLE AND DESIGNED FOR INCREASED WIND LOADING

The torsional wind load cases need not be considered if the design wind pressure in Cases 1 and 3 of Fig. 27.3-8 is increased by a factor of 1.5

D.6 CLASS 1 AND CLASS 2 SIMPLE DIAPHRAGM BUILDINGS $h \leq 160$ ft (48.8 m) MEETING THE FOLLOWING REQUIREMENTS (REFER TO SECTION 27.5.2)

D.6.1 Case A—Class 1 and Class 2 Buildings. Square buildings with $L/B = 1.0$ are exempted, where all the following conditions are satisfied:

1. The combined stiffness of the MWFRS in each principal axis direction shall be equal, and
2. The individual stiffness of each MWFRS in each principal axis direction shall be equal and symmetrically placed about the center of application of the wind load along the principal axis under consideration, and
3. The combined stiffness of the two most separated lines of the MWFRS in each principal axis direction shall be 100% of the total stiffness in each principal axis direction, and
4. The distance between the two most separated lines of the MWFRS in each principal axis direction shall be at least 45% of the effective building width perpendicular to the axis under consideration.

D.6.2 Case B—Class 1 and Class 2 Buildings. Square buildings with $L/B = 1.0$ are exempted, where all the following conditions are satisfied:

1. The combined stiffness of the MWFRS in each principal axis direction shall be equal, and
2. The individual stiffness of the two most separated lines of the MWFRS in each principal axis direction shall be equal with all lines of the MWFRS symmetrically placed about the center of application of the wind load along the principal axis under consideration, and
3. The combined stiffness of the two most separated lines of the MWFRS in each principal axis direction shall be at

least 66% of the total stiffness in each principal axis direction, and

4. The distance between the two most separated lines of the MWFRS in each principal axis direction shall be at least 66% of the effective building width perpendicular to the axis under consideration.

D.6.3 Case C—Class 1 and Class 2 Buildings. Rectangular buildings with L/B equal to 0.5 or 2.0 ($L/B = 0.5$, $L/B = 2.0$) are exempted, where all the following conditions are satisfied:

1. The combined stiffness of the MWFRS in each principal axis direction shall be proportional to the width of the sides perpendicular to the axis under consideration, and
2. The individual stiffness of each of the MWFRS in each principal axis direction shall be equal and symmetrically placed about the center of application of the wind load along the principal axis under consideration, and
3. The combined stiffness of the two most separated lines of the MWFRS in each principal axis direction shall be 100% of the total stiffness in each principal axis direction, and
4. The distance between the two most separated lines of the MWFRS in each principal axis direction shall be at least

80% of the effective building width perpendicular to the axis under consideration.

D.6.4 Case D—Class 1 and Class 2 Buildings. Rectangular buildings with L/B equal to 0.5 or 2.0 ($L/B = 0.5$, $L/B = 2.0$) are exempted, where all the following conditions are satisfied:

1. The combined stiffness of the MWFRS in each principal axis direction shall be proportional to the width of the sides perpendicular to the axis under consideration, and
2. The individual stiffness of the most separated lines of the MWFRS in each principal axis direction shall be equal with all lines of the MWFRS symmetrically placed about the center of application of the wind load along the principal axis under consideration, and
3. The combined stiffness of the two most separated lines of the MWFRS in each principal axis direction shall be at least 80% of the total stiffness in each principal axis direction, and
4. The distance between the two most separated lines of the MWFRS in each principal axis direction shall be 100% of the effective building width perpendicular to the axis under consideration.

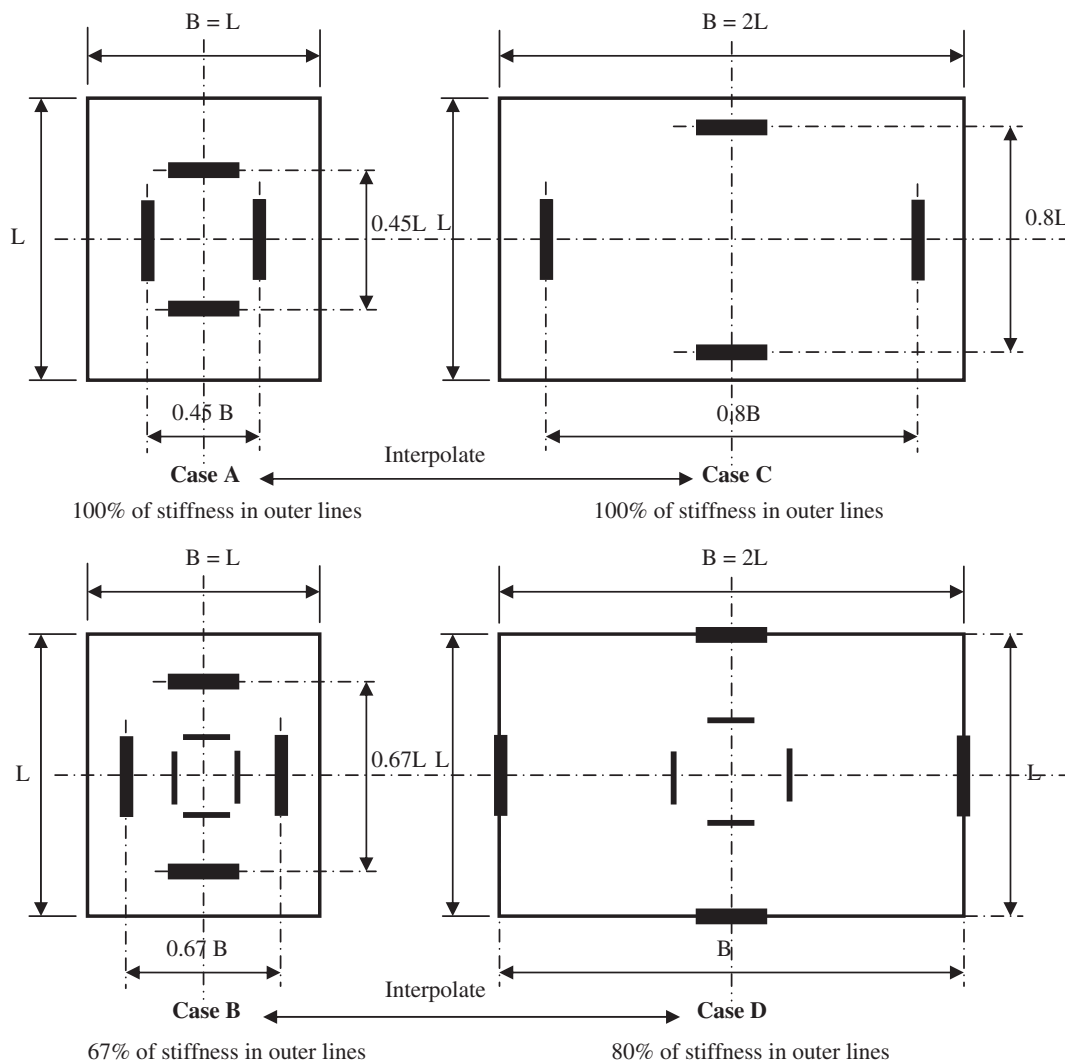
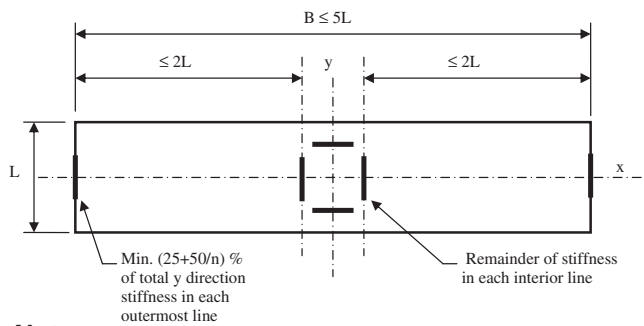


Figure D.6-1 Main Wind Force Resisting System, Appendix D ($h \leq 160$ ft) (48.8 m): MWFRS Requirements of Case E for Enclosed Simple Diaphragm Buildings. For Wind Torsion Exclusion, see Figure 27.3-8.



Note

n = required number of lines of resistance in each principal axis direction ($2 \leq n \leq 4$)

Figure D.6-2 Main Wind Force Resisting System, Appendix D ($h \leq 160$ ft) (48.8 m): MWFRS Requirements of Case F for Enclosed Simple Diaphragm Buildings. For Wind Torsion Exclusion, see Figure 27.3-8.

D.6.5 Case E—Class 1 and Class 2 Buildings. Rectangular buildings having L/B between 0.5 and 1.0 ($0.5 < L/B < 1.0$) or between 1.0 and 2.0 ($1.0 < L/B < 2.0$), the stiffness requirements and the separation distances between the two most separated

lines of the MWFRS in each direction shall be interpolated between Case A and Case C and between Case B and Case D, respectively (Fig. D.6-1).

D.6.6 Case F—Class 1 Buildings. Rectangular buildings having L/B between 0.5 and 0.2 ($0.2 \leq L/B < 0.5$) or between 2.0 and 5.0 ($2.0 < L/B \leq 5.0$) are exempted, see Fig. D.6-2, where all of the following conditions are satisfied:

1. There shall be at least two lines of resistance in each principal axis direction, and
2. All lines of the MWFRS shall be symmetrically placed about the center of application of the wind load along the principal axis under consideration, and
3. The distance between each line of resistance of the MWFRS in the principal axis direction shall not exceed 2 times the least effective building width in a principal axis direction, and
4. The individual stiffness of the most separated lines of the MWFRS in each principal axis direction shall be equal and not less than $(25+50/n)\%$ percent of the total stiffness where n is the required number of lines of resistance in the principal axis direction as required by conditions 1 and 3 of this section. The value of n shall be 2, 3, or 4.

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